THE CASE FOR KNOWLEDGE COMPRESSION

Knowledge retrieval and augmentation have also been shown to benefit significantly from compression. Results will vary by use case, but generally, the more verbose the source of knowledge, the more it will benefit from compression. Documents that feature literary prose, such as stories and novels, will benefit more than, say, a base of code. However, if the code is likewise very repetitive, compression could also be shown to be beneficial.

THE CASE FOR HOW OFTEN YOU APPLY COMPRESSION

Memory will often benefit from the periodic compression application, whereas knowledge stores typically only help on the first load. How frequently you apply compression will greatly depend on the memory use, frequency, and quantity.

THE CASE FOR APPLYING COMPRESSION MORE THAN ONCE

Multiple passes of compression at the same time has been shown to improve retrieval performance. Other patterns have also suggested using memory or knowledge at various levels of compression. For example, a knowledge store is compressed two times, resulting in three different levels of knowledge.

THE CASE FOR BLENDING KNOWLEDGE AND MEMORY COMPRESSION

If a system is specialized to a particular source of knowledge and that system also employs memories, there may be further optimization to consolidate stores. Another approach is to populate memory with the starting knowledge of a document directly.